

Chapter 13 –

Magnetic Effects of Electric Current

1. The magnetic field lines inside a long straight solenoid carrying current are:

- A) Circular
- B) Curved
- C) Straight and parallel
- D) Irregular

Answer: C) Straight and parallel

2. The direction of magnetic field around a current-carrying straight conductor is given by:

- A) Fleming's left-hand rule
- B) Fleming's right-hand rule
- C) Right-hand thumb rule
- D) Ampere's law

Answer: C) Right-hand thumb rule

3. The region around a magnet where its influence can be felt is called:

- A) Magnetic line
- B) Magnetic field
- C) Magnetic pole
- D) Magnetic force

Answer: B) Magnetic field

4. Magnetic field lines emerge from:

- A) South pole to north pole
- B) North pole to south pole
- C) Both directions
- D) None

Answer: B) North pole to south pole (outside the magnet)

5. Inside the magnet, magnetic field lines move from:

- A) South to north
- B) North to south
- C) Randomly
- D) Both ways

Answer: A) South to north

6. The magnetic field produced by a current-carrying wire depends on:

- A) Current
- B) Distance from wire
- C) Medium
- D) All of these

Answer: D) All of these

7. The magnetic field at the center of a circular coil increases if:

- A) Current decreases
- B) Radius increases
- C) Number of turns increases
- D) Coil is made smaller

Answer: C) Number of turns increases

8. The north pole of a freely suspended magnet points towards:

- A) Geographical north
 - B) Geographical south
 - C) Magnetic south
 - D) Magnetic north
- Answer: A) Geographical north
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9. A current-carrying conductor placed in a magnetic field experiences:

- A) Electric force
- B) Magnetic force
- C) Gravitational force
- D) No force

Answer: B) Magnetic force

10. The direction of force on a current-carrying conductor in a magnetic field is given by:

- A) Right-hand thumb rule
- B) Fleming's left-hand rule
- C) Fleming's right-hand rule
- D) Ampere's law

Answer: B) Fleming's left-hand rule

11. The direction of induced current is given by:

- A) Fleming's left-hand rule
- B) Fleming's right-hand rule
- C) Lenz's law

D) Both B and C

Answer: D) Both B and C

12. The magnetic field strength inside a solenoid depends on:

A) Current

B) Number of turns per unit length

C) Core material

D) All of these

Answer: D) All of these

13. A solenoid behaves like:

A) Bar magnet

B) Circular loop

C) Straight conductor

D) Electroscope

Answer: A) Bar magnet

14. The magnetic effect of current was discovered by:

A) Faraday

B) Oersted

C) Ampere

D) Maxwell

Answer: B) Oersted

15. The right-hand thumb rule gives:

A) Direction of current

B) Direction of magnetic field

C) Direction of force

D) None

Answer: B) Direction of magnetic field

16. An electric motor converts:

A) Electrical energy into mechanical energy

B) Mechanical energy into electrical energy

C) Electrical into light

D) None

Answer: A) Electrical energy into mechanical energy

17. An electric generator converts:

- A) Electrical energy into mechanical energy**
- B) Mechanical energy into electrical energy**
- C) Electrical into light**
- D) Electrical into heat**

Answer: B) Mechanical energy into electrical energy

18. The working principle of an electric generator is based on:

- A) Heating effect**
- B) Electromagnetic induction**
- C) Magnetic effect**
- D) Chemical effect**

Answer: B) Electromagnetic induction

19. The device used to convert mechanical energy into electrical energy is:

- A) Motor**
- B) Generator**
- C) Battery**
- D) Transformer**

Answer: B) Generator

20. The direction of induced current in a coil is such that it opposes the cause producing it. This is known as:

- A) Fleming's rule**
- B) Lenz's law**
- C) Ampere's rule**
- D) Faraday's law**

Answer: B) Lenz's law

21. The phenomenon of producing current by changing magnetic field is called:

- A) Electromagnetism**
- B) Electromagnetic induction**
- C) Electric effect**
- D) Magnetic resonance**

Answer: B) Electromagnetic induction

22. In an AC generator, current changes direction:

- A) 10 times per second**
- B) 50 times per second**

C) 100 times per second

D) 60 times per second

Answer: C) 100 times per second (because frequency = 50 Hz)

23. The device used to increase or decrease AC voltage is called:

A) Generator

B) Motor

C) Transformer

D) Rectifier

Answer: C) Transformer

24. The magnetic field inside a long solenoid is:

A) Zero

B) Uniform

C) Non-uniform

D) Decreasing

Answer: B) Uniform

25. The energy losses in an electric motor are mainly due to:

A) Friction

B) Heat

C) Resistance of coil

D) All of these

Answer: D) All of these

26. The type of current supplied to our homes is:

A) DC

B) AC

C) Both

D) Static

Answer: B) AC

27. The direction of current in DC:

A) Changes continuously

B) Changes periodically

C) Remains same

D) Alternates 50 times

Answer: C) Remains same

28. The symbol of an electric motor is represented by:

- A) M
- B) G
- C) T
- D) E

Answer: A) M

29. The commercial frequency of AC in India is:

- A) 60 Hz
- B) 100 Hz
- C) 50 Hz
- D) 25 Hz

Answer: C) 50 Hz

30. The domestic electric circuit works on:

- A) DC only
- B) AC of 220 V
- C) AC of 110 V
- D) DC of 220 V

Answer: B) AC of 220 V
